

1. System Architectural Overview

VectorEdge replaces centralized video streaming with distributed edge state estimation.

Traditional approaches stream full-bandwidth H.264/H.265 video over wide-area networks to centralized cloud analytics. This model fails in DDIL (Denied, Disrupted, Intermittent, Limited) environments due to bandwidth exhaustion, egress expense, and latency spikes.

VectorEdge deploys a localized compute agent adjacent to the sensor array. High-dimensional visual and acoustic frames are ingested, converted into low-dimensional kinematic state vectors at line rate, and evaluated via lightweight statistical classifiers directly on local silicon.

2. Data Pipeline: Ingestion to Vectorization

The edge runtime follows a four-stage sequential pipeline:

Stage A: Ingestion

- Local RTSP/ONVIF streams ingested via zero-copy memory buffers.
- Target hardware: Low-power edge NPUs / ARM Cortex-A processors (15–50 BOM).

Stage B: Local Feature Extraction

- Raw video frames are stripped down to non-invertible mathematical primitives:
 - Centroid Velocity (vx, vy, vz)
 - Spatial Trajectory Curvature (dtheta / dt)
 - Normalized Bounding Box Aspect Ratio (w / h)
 - Calibrated Acoustic Energy Bands (dB level across 3 frequency bands)
 - Optical Thermal Footprint Area (sq meters equivalent)

Stage C: Additive State Classification

- Local state evaluation via 1D additive log-odds scoring (Naive Bayes / LDA):

$$\log\left(\frac{\Pr(\text{Safe} \mid X)}{\Pr(\text{Threat} \mid X)}\right) = a_k + \sum_{j=1}^p g_{\{k\}}(x_j)$$

- Eliminates multi-dimensional matrix inversions and tensor overhead.
- Execution time: < 8 milliseconds per frame on embedded edge silicon.

Stage D: Telemetry Transmission

- Payload serialization: Compact Protobuf / binary schema (< 256 bytes per event).
- Transport protocols: CoAP, MQTT over TLS, or raw UDP burst over tactical radios.
- Egress requirement: 64 kbps to 128 kbps aggregate per 50-camera sector.

3. Bandwidth and Compute Comparison

Raw Video Streaming (Industry Standard)

- Network Bandwidth: 2.5 Mbps to 8.0 Mbps per camera.
- Uplink Requirement: High-speed commercial fiber or unthrottled LTE/5G.
- Incident Detection Latency: 3,000 ms to 6,000 ms (round-trip cloud pipeline).
- Failure State in Contested Links: Complete video loss, zero situational awareness.

VectorEdge Telemetry Ingestion

- Network Bandwidth: < 1.0 KB/sec per camera zone.
- Uplink Requirement: Degraded cellular (2G/3G), HF tactical radio, or low-tier SATCOM.
- Incident Detection Latency: < 10 ms (evaluated locally at the edge).
- Failure State in Contested Links: Local physical interlocks execute autonomously; telemetry queues in local ring buffer until uplink restores.

4. Threat Classification Matrix

Class 0: Routine Ambient Traffic (Authorized Fleet / Fauna)

- Execution: Vector logged locally, suppressed from uplink to conserve bandwidth.

Class 1: Operational Anomaly (Loitering / Off-Route Movement)

- Execution: Low-bandwidth vector burst forwarded to operations center.

Class 2: Kinetic Breach / Perimeter Compromise

- Execution: Local GPIO triggers physical relays (bollards, locks); vector broadcast over mesh; sub-frame forensic snapshot prioritized for uplink.